

Executive Summary - Iris Classification Project

Objective:

The goal of this project is to build a machine learning model that can classify iris flowers into three species (Setosa, Versicolor, Virginica) based on sepal and petal measurements.

Approach:

1. Loaded the Iris dataset and performed exploratory data analysis (EDA).
2. Visualized relationships using pairplots and boxplots.
3. Split the dataset into training and testing sets.
4. Trained three models: K-Nearest Neighbors (KNN), Logistic Regression, and Decision Tree.
5. Compared model performance using accuracy, confusion matrix, and classification report.
6. Selected the best-performing model and saved it as iris_model.pkl.

Results:

- All models achieved high accuracy (>90%).
- The best model was identified based on test accuracy and confusion matrix results.
- The saved model successfully predicts species for new samples.

Conclusion:

The Iris Classification project demonstrates the complete machine learning workflow: data preparation, visualization, model training, evaluation, and deployment. The final model is reliable and can be extended for real-world applications such as automated plant classification systems.